

A chemistry-stratified reliability audit of selective abstention for universal machine-learning interatomic potentials: abstention value is oxide-anchored and control-hardened across 32 models

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Universal machine-learning interatomic potentials (UIPs) disagree most with the MP2020-corrected DFT stability labels they are scored against in oxide chemistries — formation-energy RMSE $\sim 7\times$ that of other anion classes on Matbench Discovery, consistent with oxides being where semilocal-DFT (GGA) energetics, the Hubbard- U treatment they lack, and the layered MP2020 corrections are least mutually consistent. The value of selective abstention in UIPs is therefore chemistry-dependent and, in our central finding, oxide-anchored: a genuine +7.7 precision-point advantage (CHGNet oxide-vs-halide) that we establish with a reusable, CPU-only matched-yield audit protocol, runnable on any published prediction file. Across six anion-class strata and four UIPs on Matbench Discovery, the stratum-by-coverage interaction in discovery acceleration factor (DAF) excludes zero in 45 of 60 cells under a prototype-blocked bootstrap (48/60 i.i.d.), is oxide-anchored, and survives leave-one-element-out, WBM-round, multiplicity, and alternative-stratification controls with a shuffle null. Low-base-rate normalisation amplifies this to the +0.90 DAF headline (95 % CI [0.74, 1.04]); the precision-point interaction is the primary effect size throughout. Across 44 published-potential files (43 independent models), the oxide anchor holds for all 32 UIPs with no detectable generational difference ($p=0.68$, an honest null). The established finding is thus that abstention value — and the UIP-label disagreement driving it — is chemistry-localised, anchored in the oxide stratum where the reference labels' construction is least self-consistent. One interpretation, which our disagreement metric cannot isolate from UIP-fitting difficulty, is that this precision headroom co-locates with where the benchmark's MP2020-corrected labels are least reliable — a reference-data-trustworthiness reading we flag but do not establish. Scoping the claim: the oxide-highest ordering does not transfer temporally, and a stratum-aware allocator is falsified.

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I. INTRODUCTION

Evaluating the *reliability* of universal machine-learning interatomic potentials (UIPs) — not merely their aggregate accuracy, but when their predictions can be trusted enough to act on — has become central to using them for materials screening. Our central finding is a chemical-physics one: the value of confidence-based selective abstention in UIP stability screening is not uniform across chemistry, and it is largest of all in oxides — consistent with oxides being the chemistry where the semilocal-DFT (GGA) reference energetics, the correlated-*d* Hubbard-*U* treatment they lack, and the composition- and oxidation-state-dependent MP2020 corrections layered on top of them are least mutually self-consistent (see Discussion, “Why UIP-label disagreement is largest in oxides”). One candidate explanation for this co-location, which we flag but do not establish, is that the chemistry-stratified abstention interaction signals where the DFT reference data underlying UIP training targets is least trustworthy; either way it is not a deployable stratified abstention policy — a reading we are careful not to overstate, since the UIP-fitting and label-reliability explanations co-locate in oxides without being separated here. Its honest nulls — a clean shuffle control, no detectable generational difference, a falsified stratum-aware allocator, and non-transfer under a temporal hold-out — are not caveats bolted on afterward but the very evidence that fixes this diagnostic (not allocator) reading: they are the protocol’s rigor made visible. Matbench Discovery¹ established UIPs as the standard first-pass filter for thermodynamic stability: given a relaxed candidate structure, a UIP predicts its formation energy, from which one computes a predicted distance to the convex hull and labels the structure stable when that distance is below zero. The benchmark’s WBM test set² contains ~257k substitution-generated structures with corrected DFT ground truth, and models such as CHGNet³, M3GNet⁴, MACE⁵, and ORB⁶ now achieve useful stable/unstable discrimination across it.

A practitioner rarely needs to label every candidate. The operational question is selection: which subset should be carried forward to expensive DFT or synthesis? Selective abstention — keeping only the highest-confidence predicted-stable structures and declining to commit on the rest — is the natural mechanism, and the absolute predicted hull margin $|\hat{e}_{\text{hull}}|$ is a free, model-native confidence signal. We establish the finding above with a reusable, CPU-only matched-yield audit protocol, runnable on any set of already-published UIP prediction files: it needs no retraining or GPU, and gates every reported number to a frozen, re-runnable result. The oxide-anchored effect we report on Matbench Discovery is the worked demonstration of that protocol. The relevant figure of merit

is the discovery acceleration factor (DAF), the precision of the surfaced shortlist divided by the stratum’s stable base rate, which is comparable across chemistries with different base rates.⁷ The closest prior work on certifiable abstention for UIPs, PCM⁸, reports a global risk AUC-ROC of 0.938 but does not stratify the value of abstention by anion chemistry or apply a matched-yield control — those are the contributions of the present study.

A companion selective-prediction study tested whether other confidence signals — isotonic-calibrated p_{stable} and cross-model ensemble disagreement — beat the free hull-margin baseline at matched coverage and matched stable-call yield, and found that none did (0/12 model $\times$ coverage cells): hull margin itself was undefeated, i.e. “distance is all you need” for this task. That result killed the wedge of a learned/calibrated signal beating margin, not margin-based abstention itself, and motivates a complementary, conditional question. Hull margin was shown to be a sufficient, undefeated abstention signal in aggregate — but is its benefit nonetheless chemistry-dependent? If abstention buys far more in some anion classes than in others at a matched surfaced-candidate budget, then a stratified abstention policy sharpens an already-sufficient global one.

We contribute a reusable, matched-yield reliability-audit protocol for UIP abstention and apply it to Matbench Discovery. Concretely, this paper makes three contributions.

- We define genuine anion-class strata (electronegativity-priority assignment per formula) with materially different stable base rates (oxide 12.4 % vs. halide 27.8 %) and measure the stratum $\times$ coverage interaction in DAF under a *common-absolute-yield* control that reproduces the exact matched-yield semantics of that companion study, lifted to strata.
- We show the interaction is real (45/60 cells exclude zero under a prototype-blocked bootstrap; 48/60 under i.i.d. resampling), directional (oxides highest, halides/intermetallics lowest), and physically coherent, with a clean within-stratum shuffle-null (0/60).
- We harden it on two independent robustness axes — leave-one-element-out (LOEO) and WBM-round (time-like) cross-splits — and report transparently where statistical power, not the effect, runs out.
- We decompose the effect into a genuine precision advantage (+7.7 precision points, CHGNet oxide-vs-halide) and a base-rate-normalization factor that DAF folds in, report the precision-point interaction as primary, and trace the precision headroom to UIP-label disagreement

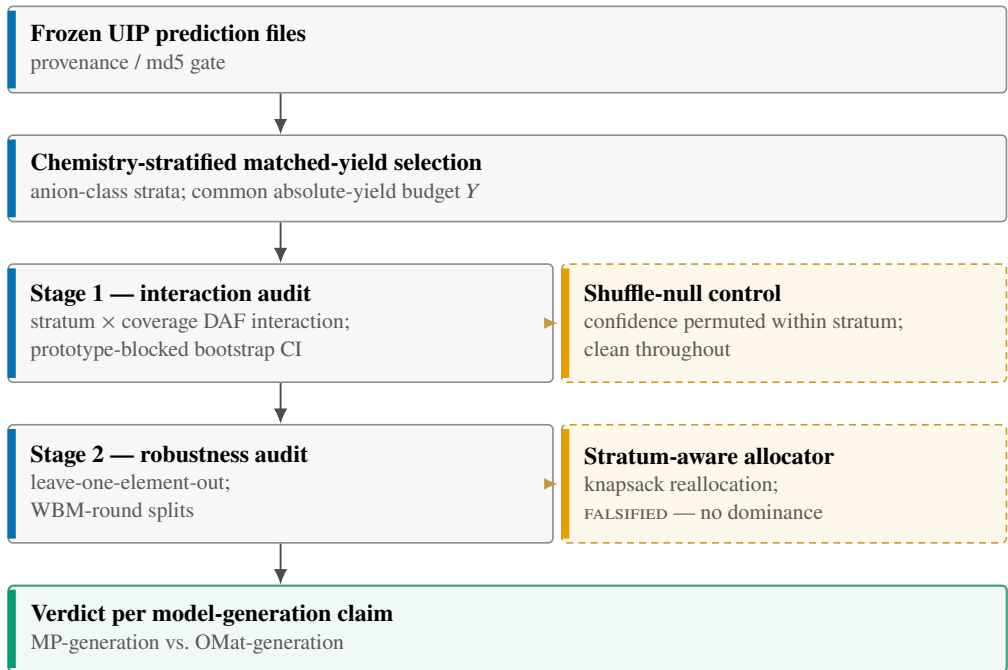


FIG. 1: The matched-yield reliability-audit protocol. Frozen, provenance/md5-gated UIP prediction files feed a chemistry-stratified matched-yield selection (anion-class strata, common absolute-yield budget Y), which is audited in two stages — the Stage 1 stratum $\times$ coverage DAF interaction and the Stage 2 leave-one-element-out / WBM-round robustness gates — to a verdict per model-generation claim. Two built-in controls (dashed) guard the audit: a shuffle-null that stays clean throughout, and a stratum-aware knapsack allocator that the protocol itself falsified.

on the benchmark’s least trustworthy slice — establishing the interaction as a diagnostic of reference-data trustworthiness.

We are explicit about scope throughout: the result is an abstention-quality interaction at matched yield, read as a reference-data-trustworthiness diagnostic and not as a deployable stratified allocator (which its own temporal-transfer and knapsack controls falsify), and not as a claim that stratification re-orders the models, which it does not. The full scope statement — what this result establishes and what remains explicitly out of scope — is given in the Discussion (“Scope and what stays dead”); we flag the boundary here, before the Results, so it need not be inferred retroactively. Figure 1 summarizes the protocol end to end, from the frozen, provenance-gated prediction files through the two audit stages to the per-model-generation verdict, with its two built-in controls — the shuffle-null and the falsified stratum-aware allocator — shown as explicit checks.

a. Notation. Table I collects the recurring short-codes used throughout, as a compact reference map; each term remains defined inline at its first use in the running text below.

TABLE I: Notation used throughout this paper. Each term is also defined inline at first use; this table is a reference map, not a substitute for those definitions.

Term	Meaning
UIP	Universal machine-learning interatomic potential
WBM	Matbench Discovery’s substitution-generated test set, with corrected DFT ground truth
DAF	Discovery acceleration factor: surfaced-shortlist precision $\div$ stratum stable base rate
$\hat{e}_{\text{hull}}$	Predicted (signed) hull distance; $ \hat{e}_{\text{hull}} $ is the absolute margin used as the confidence signal
$Y_{\text{loose}}, Y_{\text{tight}}$	Common absolute surfaced-yield budgets (looser, tighter), held identical across strata
$\text{gain}_A - \text{gain}_B$	Stratum $\times$ coverage interaction: the matched-yield abstention-gain difference between strata A, B
LOEO	Leave-one-element-out robustness cross-split
WBM round	Acquisition-round (time-like) robustness cross-split
p_{stable}	Calibrated (or native-Gaussian) probability of stability
SSCS	Worst-slab calibration gap, $\max_s \text{gap}_s $ over strata s

II. METHODS

a. Data. We use Matbench Discovery’s public, on-disk cached artifacts: the WBM summary ground truth (corrected formation energy and corrected hull distance per structure identifier and chemical formula) and one cached predicted-formation-energy CSV per UIP. After dropping rows with missing values and inner-joining all four model prediction sets, 256,963 structures remain. No network access, tokens, or downloads occur at run time; all computation is CPU/pandas. The complete matched-yield interaction (Stage 1) and its two robustness gates (Stage 2: leave-one-element-out and WBM-round, each with a 1000-sample bootstrap) reproduce byte-for-byte from the frozen prediction files in under five CPU-minutes on a commodity laptop (measured ~ 3 min, single core, no GPU). A structure is *stable* when its corrected true hull distance is below zero; the predicted hull distance is reconstructed as the true hull distance plus the model’s signed formation-energy

error (a fixed-hull approximation validated against a from-scratch phase-diagram recompute in the Discussion), and confidence is its absolute value $|\hat{e}_{\text{hull}}|$.

b. Anion-class strata. Each formula is assigned to exactly one stratum by an electronegativity-priority rule: halide > oxide > chalcogenide (S/Se/Te) > pnictide (N/P/As/Sb/Bi) > intermetallic (all-metal) > other (remaining main-group). All six strata contain at least $\sim 23\text{k}$ structures and exhibit genuinely different stable base rates, which is what makes DAF (base-rate-normalised) the appropriate cross-stratum metric.

c. Matched-yield abstention gain (the companion-study control). To eliminate the surfaced-candidate-yield confound that killed the unstratified wedge, we fix a *common absolute yield budget* Y : the same integer number of surfaced predicted-stable structures in every stratum within a split. Per model we set $Y_{\text{loose}} = \min$ over usable strata of the total called-stable count, and $Y_{\text{tight}} = Y_{\text{loose}}/2$. Within each stratum we rank called-stable structures by confidence, take the top- Y , and define the abstention gain as

$$\text{gain} = \text{DAF}(\text{top } Y_{\text{tight}}) - \text{DAF}(\text{top } Y_{\text{loose}}),$$

with Y identical across strata. Because both the surfaced count and its change are held fixed across strata, any surviving cross-stratum difference is an abstention-quality-by-chemistry effect, not a yield-count artifact.

d. Interaction, bootstrap, and null. The stratum $\times$ coverage interaction for a stratum pair (A, B) is $\text{gain}_A - \text{gain}_B$, evaluated per model. We bootstrap within strata ($n_{\text{boot}} = 1000$, seed 20260621) and report the median and the 95 % percentile CI; a cell “fires” when its CI excludes zero. A shuffle/permutation null control was committed ex ante in a dated pre-analysis design memo (written hours before Stage-1 ran): it permutes the confidence signal within each stratum’s called-stable set, so abstention selects a random top- Y and the gain should vanish; we require this shuffle-null to be clean (near-zero firing) under every split. Because WBM is generated by elemental substitution into shared prototypes, called-stable structures are not independent within a stratum; we therefore additionally compute every Stage-1 CI with a *prototype-blocked* bootstrap that resamples structural-prototype clusters with replacement within each stratum (block = WBM AFLOW prototype label), and adopt the blocked CIs as primary. All other quantities (budgets, base rates, DAF points) are identical to the i.i.d. run and reproduce it byte-for-byte.

e. Robustness gates. Two independent cross-splits harden the Stage-1 interaction. (i) *Leave-one-element-out (LOEO)*: drop every structure whose formula contains a target element E and

recompute — for the most common anion (O), the dominant cations (Ni, Al, Ge, Cu, Fe, Sn, Si), and the stratum-defining anion-formers (F, S, N, P), giving 12 splits. (ii) *WBM round*: restrict to each WBM acquisition round r (parsed from the round number embedded in each structure’s identifier), a time-like distribution-shift axis, giving 5 splits. Thin strata are guarded by minimum-row (500) and minimum-called-stable (40) thresholds and dropped from a split if below threshold. The general direction and kill criterion for this robustness gate were committed ex ante in a dated pre-analysis design memo (21:00, hours before Stage-1 ran); the exact operational SUCCESS threshold — the interaction excludes zero for a majority of cells in a majority of LOEO splits and a majority of rounds, with the shuffle-null clean throughout — is documented in a Stage-2 analysis-protocol log written in the same analysis session as the Stage-2 results it reports, not registered before Stage-1.

III. RESULTS

A. Matched-yield abstention lifts DAF most in oxides

Figure 2 shows DAF at the looser and tighter yield budgets for every (stratum, model) pair. Across models, abstaining (moving from Y_{loose} to the tighter Y_{tight}) raises DAF, but the size of that lift is strongly chemistry-dependent: the oxide bars show the largest gap between the two coverages, whereas halide and intermetallic bars barely separate. For CHGNet the oxide gain is +1.14 DAF units versus +0.24 for halides; for MACE and ORB the intermetallic gain is essentially zero (−0.004 and −0.005) while the oxide gain remains positive (+0.67 and +0.45). The lift is a genuine reordering of which chemistry benefits, at a surfaced-candidate budget held identical across strata.

B. The stratum $\times$ coverage interaction fires, oxide-anchored and positive

Figure 3 renders the matched-yield interaction $\text{gain}_A - \text{gain}_B$ for every (model, stratum-pair) cell; outlined cells have a 95 % bootstrap CI excluding zero. Of the 60 cells, 48 exclude zero under an i.i.d. bootstrap, and all four models contribute (CHGNet 13/15, MACE 13/15, M3GNet 11/15, ORB 11/15). Because WBM is generated by elemental substitution into shared prototypes, structures sharing a parent prototype are correlated; we therefore recompute the CIs with a prototype-blocked

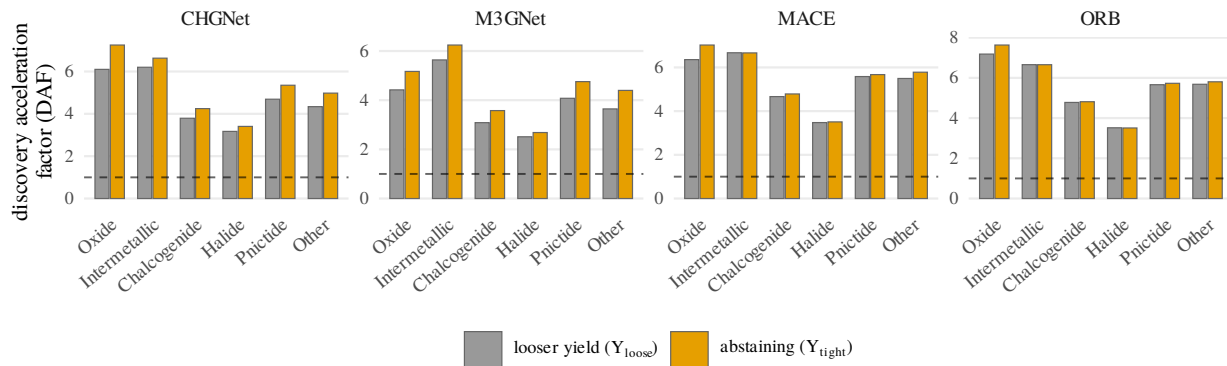


FIG. 2: Matched-yield DAF at the looser (Y_{loose}) and tighter (Y_{tight} , abstaining) yield budgets, per anion-class stratum and per model. Oxides show the largest DAF lift from abstention; halides and intermetallics the smallest.

bootstrap that resamples structural-prototype clusters within each stratum (block = WBM AFLOW prototype). Under blocking 45 of 60 cells still exclude zero (the shuffle-null stays 0/60), and we adopt this as the primary count. The three cells that drop out are all borderline intermetallic pairs whose i.i.d. CI already grazed zero (M3GNet oxide–intermetallic and intermetallic–chalcogenide, ORB intermetallic–pnictide); no oxide-vs-halide or strong oxide cell is affected. The firing cells are dominated by oxide-anchored pairs: 18 of 20 oxide-containing cells fire under i.i.d. resampling and 17 of 20 under blocking, with oxide as the higher-benefit stratum — a direction first observed at Stage-1 and documented, not registered ex ante, in a same-session Stage-2 analysis-protocol log; the general stratum-interaction hypothesis and kill criterion were committed ex ante in a dated pre-analysis design memo. The largest interaction is CHGNet’s oxide-vs-halide pair at +0.90 (prototype-blocked 95 % CI [0.74, 1.04]; i.i.d. CI [0.77, 1.04]); the median absolute interaction across all 60 cells is 0.19. These counts survive formal multiplicity control with two-sided bootstrap sign p -values per cell, Benjamini–Hochberg at $q = 0.05$ retains 46/60 cells (shuffle-null: 0/60), and a higher-resolution pass (10^4 bootstrap draws, same seed and machinery) retains 40/60 under the stricter family-wise Holm–Bonferroni procedure at $\alpha = 0.05$; the headline CHGNet oxide-vs-halide cell has Holm-adjusted $p = 0.012$. The interaction is thus empirically oxide-anchored; the next subsection dissects its mechanism directly from the predictions and finds it is driven by error-headroom and base-rate normalization, not by hull-margin confidence being intrinsically sharper in oxide chemistry.

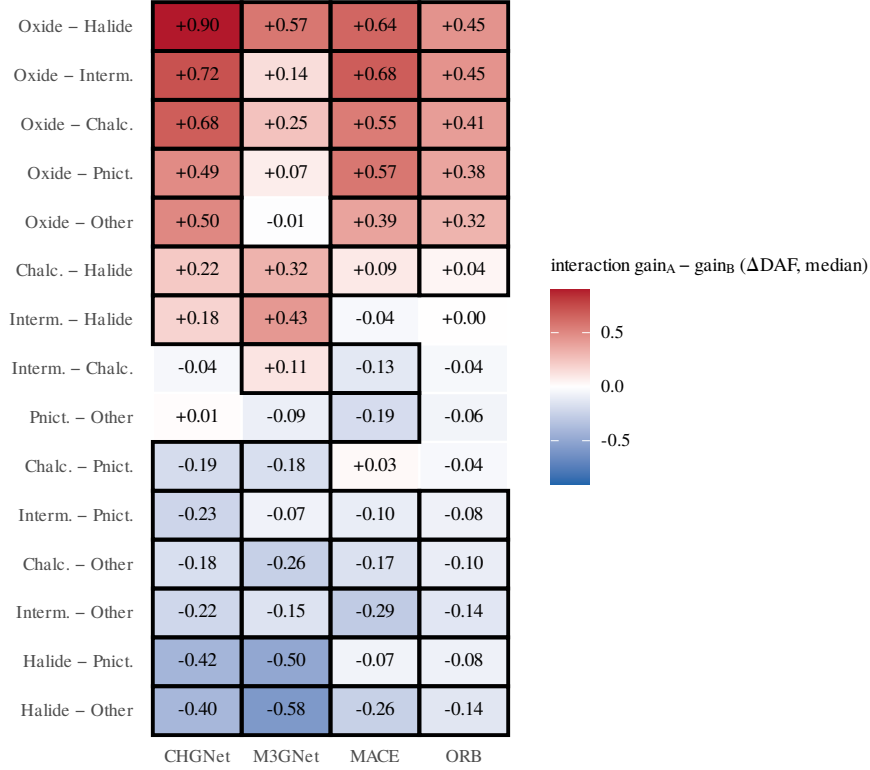


FIG. 3: Stratum $\times$ coverage interaction (median gain_A – gain_B) per model and stratum pair. Outlined cells have a 95 % bootstrap CI excluding zero (48/60 i.i.d.; 45/60 under a prototype-blocked bootstrap, the primary count — Results). Oxide-anchored pairs fire positive across models.

C. Mechanism: error headroom and base-rate normalization, not sharper confidence

To test the naive reading — that hull-margin confidence simply discriminates oxide errors better — we compute three post-hoc descriptive diagnostics per stratum directly from the cached predictions (averaged over the four UIPs in Table II; full per-model values in the reproduction archive). Two findings reframe the mechanism. First, oxides carry by far the largest UIP formation-energy error — RMSE 0.62 eV/atom versus ~ 0.09 for every other stratum — so a large share of the structures a UIP calls stable in oxides are false positives (called-stable precision 0.75), leaving room for abstention to raise precision. Second, and against the naive reading, the within-stratum rank-informativeness of the confidence signal (AUROC of $|\hat{e}_{\text{hull}}|$ separating truly-stable from false-positive called-stable structures) is if anything the lowest of the six strata in oxides (0.69 versus 0.75–0.78 elsewhere): confidence does not discriminate oxide errors better per structure. The oxide anchoring instead arises from two compounding factors — the large error creates the false-positive headroom, and

TABLE II: Post-hoc descriptive per-stratum mechanism diagnostics, averaged over the four UIPs. Oxides combine the largest formation-energy error and the lowest stable base rate with — against the naive reading — the lowest confidence-rank AUROC. Base rate is the stratum stable fraction; RMSE is the formation-energy error; called-stable precision is the true-stable fraction among structures called stable; AUROC is the rank separation of $|\hat{e}_{\text{hull}}|$ for truly-stable versus false-positive called-stable structures.

Stratum	Base rate	RMSE (eV/atom)	Called-stable prec.	Conf. AUROC
Oxide	0.124	0.625	0.747	0.695
Intermetallic	0.142	0.091	0.631	0.756
Chalcogenide	0.205	0.088	0.725	0.766
Halide	0.278	0.090	0.752	0.756
Pnictide	0.171	0.097	0.680	0.775
Other	0.167	0.097	0.641	0.772

because DAF normalizes precision by the stratum stable base rate, which is lowest in oxides (12.4 % versus 27.8 % for halides), an equal precision gain from abstention becomes a larger DAF gain. These two amplifications should be read separately, because DAF folds a genuine precision-recovery advantage together with the $1/\text{base-rate}$ factor. In precision-point units the CHGNet oxide-vs-halide abstention advantage is +7.7 points (oxide called-stable precision lifts +14.3 points under abstention, halide only +6.6); the headline +0.90 DAF interaction is that $2.2\times$ precision-recovery advantage further multiplied by DAF’s $2.2\times$ base-rate denominator (halide 27.8 % / oxide 12.4 %), so the $\sim 4.9\times$ cross-stratum DAF-gain ratio is roughly half genuine precision headroom and half base-rate normalization. We report the precision-unit interaction alongside DAF throughout so the base-rate amplification is transparent. The effect is real and reproducible — it is exactly what the shuffle-null, LOEO, round, and alternative-stratification controls certify — but it is an interaction between error-driven headroom and base-rate-normalized reporting, not evidence that a UIP is more trustworthy when confident in oxide chemistry specifically.

a. What the interaction buys a practitioner. Concretely, for CHGNet — the strongest firer of the four UIPs, so this is a best-case illustration rather than a typical one — the +0.90 oxide-versus-halide interaction converts as follows, on the *pooled* evaluation (post-hoc descriptive, from the frozen matched-yield DAF; this pooled framing is the one that does not transfer temporally,

per Threats to validity). Of the 2,696 structures CHGNet calls stable in oxides, keeping only the 1,348 most confident by $|\hat{e}_{\text{hull}}|$ raises true-stable precision from 75.8 % to 90.1 % (DAF 6.1 $\rightarrow$ 7.2). A practitioner with a fixed budget of 1,348 DFT relaxations therefore wastes ~ 134 of them on false positives by spending it on the confident half, versus ~ 326 by spending it on a same-size random draw from all called-stable oxides — roughly 190 fewer wasted relaxations. The identical abstention on halides moves precision only 88.2 % $\rightarrow$ 94.8 % (DAF 3.2 $\rightarrow$ 3.4; halide base DAF 3.01, cf. the provenance footnote), saving ~ 90 relaxations — about half the oxide payoff, which is the concrete practitioner form of the interaction.

D. The interaction is robust to held-out chemistry and acquisition round

Figure 4 reports the fraction of firing cells under each robustness split. Under LOEO, all 12 held-out-element splits remain majority-firing (12/12), pooling to 548/700 cells (538/700 under the prototype-blocked bootstrap, all 12 splits still majority-firing); crucially, dropping oxygen — which empties the entire oxide stratum and leaves only 5 strata (40 cells) — still fires 37/40 (36/40 blocked), so the effect is not carried by oxides alone, and no single cation or anion-former split falls below 39/60. Under the WBM-round gate, 3 of 5 rounds are majority-firing (rounds 1–3: 43/34/43 of 60), pooling to 157/300 cells (151/300 blocked, the same 3 rounds majority-firing). Rounds 4 and 5 fall below majority (26 and 11 of 60); these are the two smallest rounds (40k and 23k rows), so the shortfall is a per-model power effect under tiny matched-yield budgets, not a temporal sign flip — the oxide-highest direction is preserved even where the CIs widen. The majority-of-splits / majority-of-rounds success criterion, documented alongside the Stage-2 results themselves rather than registered before Stage-1, is met on both axes.

E. The shuffle-null is clean everywhere

Figure 5 contrasts the magnitude of the real interaction with the confidence-shuffled interaction, per gate. Permuting confidence within each stratum’s called-stable set collapses the signal to a tight band around zero, and no shuffled cell’s CI excludes zero in any gate: 0/60 at Stage-1, 0/700 under LOEO, and 0/300 across rounds. The largest shuffled Stage-1 interaction is 0.0047 in absolute value, two orders of magnitude below the real median. The interaction is therefore a property of the

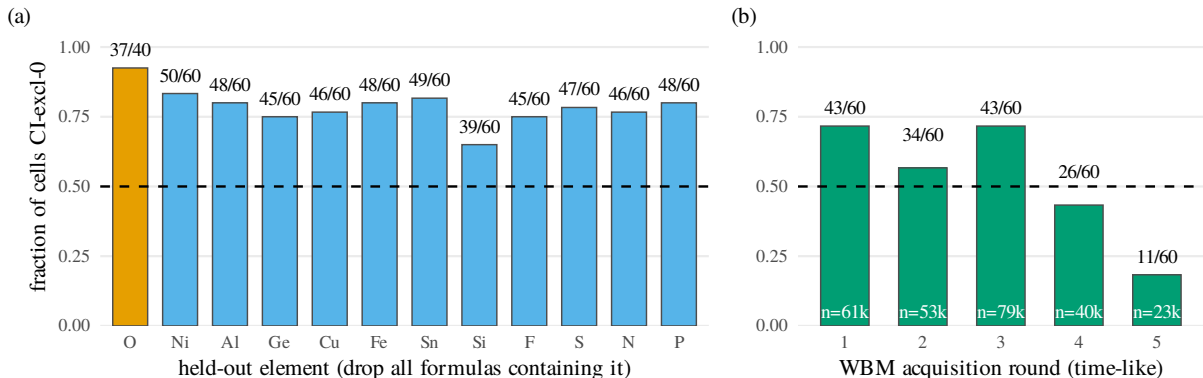


FIG. 4: Fraction of (model $\times$ stratum-pair) cells whose 95 % CI excludes zero, per robustness split. (a) leave-one-element-out (12/12 splits majority-firing); (b) WBM round (3/5 rounds majority-firing). Cell counts annotated; dashed line is the majority threshold.

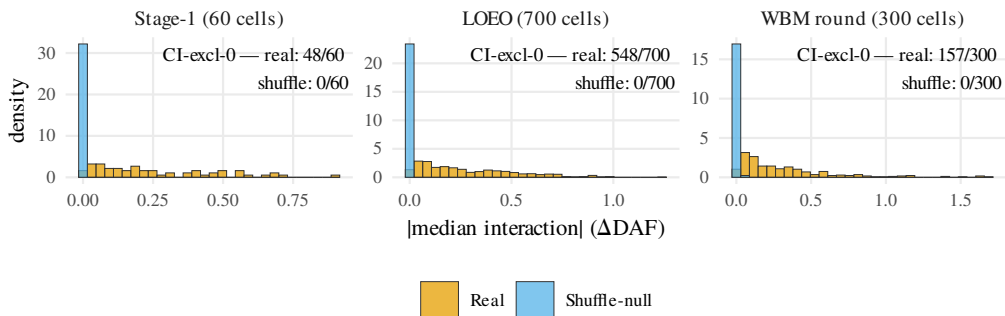


FIG. 5: Real vs. within-stratum confidence-shuffled $|\text{interaction}|$ per gate. The shuffle-null collapses to zero (0/60, 0/700, 0/300 cells with CI excluding zero).

confidence ordering, exactly what a genuine abstention effect requires.

F. Committee-variance abstention reproduces the interaction; native-Gaussian coverage is optimistic in every chemistry

Two further analyses probe whether the oxide-anchored abstention benefit is a quirk of the particular confidence source. First, we replace each UIP’s single-model $|\hat{e}_{\text{hull}}|$ with a *committee-variance* signal: confidence = $-\text{std}$ of the four UIPs’ reconstructed predicted hull distances, with the consensus stable call taken as the committee-mean hull below zero, holding the matched-yield control identical ($Y_{\text{tight}} = 1422$, $Y_{\text{loose}} = 2844$). Figure 6(a) shows the committee-variance

TABLE III: Abstention-signal comparison: the free single-model hull margin $|\hat{e}_{\text{hull}}|$ vs. a four-UIP committee-variance signal, both under the identical matched-yield DAF control. Denominators differ because the committee collapses the model axis: single-model interaction cells are counted per (model $\times$ stratum-pair) = 60, committee interactions per stratum-pair = 15. Both counts here use the i.i.d. bootstrap for an apples-to-apples comparison; the single-model primary count under the prototype-blocked bootstrap is 45/60 (Results).

	Single-model $ \hat{e}_{\text{hull}} $	Committee variance
Confidence signal	$ \hat{e}_{\text{hull}} $ per UIP	–std over 4 UIP hulls
Stable call	predicted hull < 0	committee-mean hull < 0
Matched-yield budget ($Y_{\text{tight}}/Y_{\text{loose}}$)	per model	1422/2844
Interaction cells CI-exclude-0	48/60	7/15
Shuffle-null cells CI-exclude-0	0/60	0/15
Dominant interaction sign	+ (oxide-anchored)	+ (oxide-anchored)
Sign agreement, jointly-sig. pairs	—	7/7
Robustness gate verdict	—	PASS-BOTH

abstention benefit is again concentrated in oxides (matched-yield DAF gain median +0.563) and small elsewhere ($\leq +0.10$; halide -0.014). Of the 15 stratum-pair interactions, 7 have a 95 % bootstrap CI excluding zero (Figure 6(b)): all five oxide-anchored pairs positive (oxide-vs-halide +0.574, CI [0.463, 0.687]) plus chalcogenide-vs-halide (+0.115, CI [0.044, 0.183]), with one negative (halide-vs-pnictide -0.114 , CI $[-0.211, -0.019]$), while the within-stratum confidence shuffle fires 0/15. The committee’s dominant interaction sign is positive, matching the single-model reference, and it agrees in sign on all 7 jointly-significant pairs, yet it fires far fewer pairs than the single-model consensus — so it is a genuinely distinct re-derivation of the effect, not a relabelling of one model. One limitation qualifies this as *corroboration* rather than independent confirmation: the committee is built from the same four UIPs’ predictions, so if those models share a correlated oxide error (plausibly through the common MP2020-corrected labels they are trained and scored against; see Discussion), the committee-variance signal inherits that oxide false-positive richness rather than independently confirming it. It re-derives the interaction from a different *functional* of the same predictions, not from independent models (robustness gate verdict PASS-BOTH; Table III).

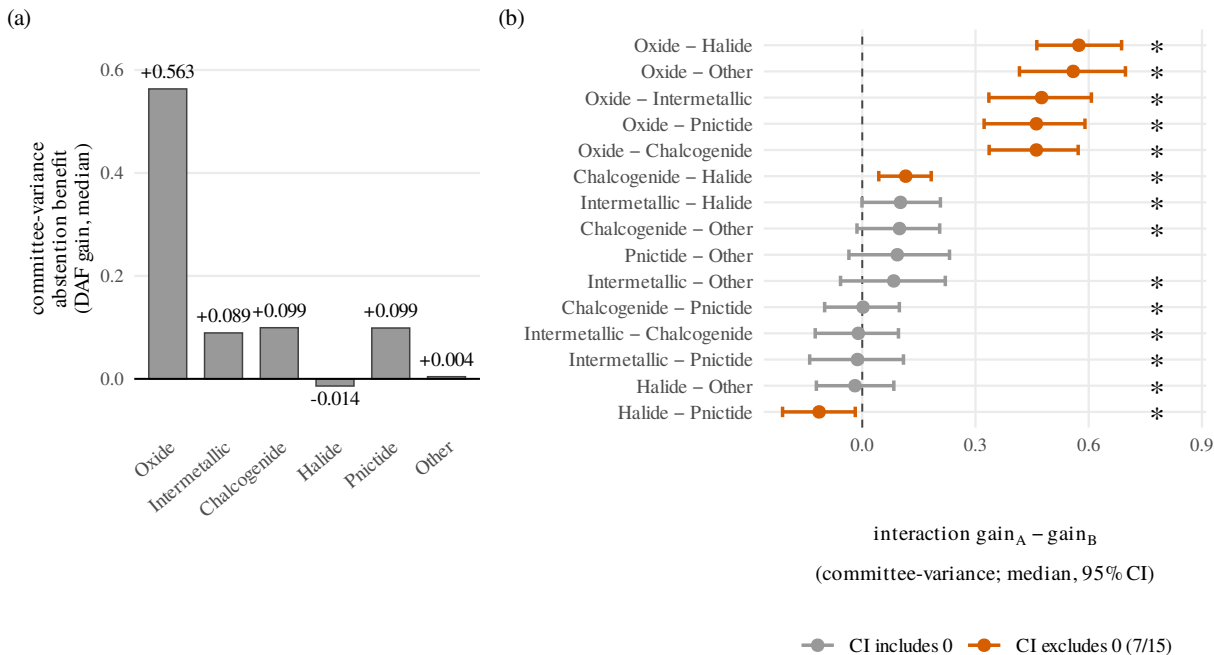


FIG. 6: Committee-variance abstention vs. the single-model signal. (a) Matched-yield DAF gain (median) per anion stratum using the committee-variance signal; the benefit concentrates in oxides, mirroring the single-model pattern. (b) The 15 stratum-pair committee-variance interactions ($\text{gain}_A - \text{gain}_B$, median with 95 % bootstrap CI); orange marks the 7 pairs whose committee CI excludes zero, and the side asterisk marks pairs on which the single-model consensus also fired. The committee is sparser (7/15) but sign-consistent; the confidence-shuffle collapses to 0/15.

Second, we ask whether a zero-free-parameter probabilistic reading of the same margin is well-calibrated within each chemistry. Using the native Gaussian $p_{\text{stable}} = \Phi(-\hat{e}_{\text{hull}}/\sigma)$, with σ the calibration-half RMSE of the formation-energy error, we measure the per-anion-family conditional-coverage gap (mean p_{native} minus the empirical stable rate) on the held-out test half. Every gap is positive and CI-excludes-zero for every UIP (Table IV): the native Gaussian over-predicts stability in every family. The worst-slab gap ($\text{SSCS} = \max_s |\text{gap}_s|$) ranges from ORB’s 0.077 (best calibrated) to MACE’s 0.322 (worst, driven by its inflated $\sigma = 0.977$, which pushes p toward 0.5 everywhere), with a cross-model mean of 0.197. Two distinct chemistry dependences must not be conflated here. Calibration miscalibration (whether p_{stable} matches the empirical stable rate) and abstention headroom (whether ranking by $|\hat{e}_{\text{hull}}|$ can lift precision) are different properties: the native Gaussian is optimistic in every family, but the stratum where it fails hardest is intermetallics for three of

TABLE IV: Native-Gaussian ($p_{\text{stable}} = \Phi(-\hat{e}_{\text{hull}}/\sigma)$, zero free parameters) per-anion-family conditional-coverage gap (mean p_{native} minus empirical stable rate) on the test half, per UIP. Every gap is positive and its 95 % bootstrap CI excludes zero, so the native Gaussian over-predicts stability in every family for every model. SSCS is the worst-slab $\max_s |\text{gap}_s|$.

UIP	σ_{cal}	Oxide	Interm.	Chalc.	Halide	Pnict.	Other	SSCS (worst)
CHGNet	0.102	0.109	0.177	0.139	0.126	0.148	0.161	0.177 (interm.)
M3GNet	0.117	0.172	0.214	0.200	0.177	0.195	0.207	0.214 (interm.)
MACE	0.977	0.322	0.315	0.246	0.199	0.277	0.287	0.322 (oxide)
ORB	0.081	0.074	0.077	0.048	0.055	0.038	0.052	0.077 (interm.)
Mean SSCS over models								0.197

the four UIPs (CHGNet, M3GNet, ORB; Table IV), not oxides, and only MACE has its worst calibration gap in oxides. The abstention benefit, by contrast, is oxide-anchored for all four models. The two axes therefore co-locate in oxides only for MACE; in general the chemistry a UIP is least calibrated on is not the chemistry where abstention pays most.

IV. DISCUSSION

a. What the result means. The value of selective abstention in UIP stability screening is not uniform across chemistry, and where it is largest tracks where the reference data is least trustworthy. At a surfaced-candidate budget held identical across anion classes, abstaining on low-confidence calls accelerates discovery substantially more in oxides than in halides or intermetallics — but the load-bearing quantity is a genuine precision advantage, not the base-rate-amplified DAF headline. Decomposed (Results, Table II), the CHGNet oxide-vs-halide abstention gain is +7.7 precision points; DAF’s normalization by the low oxide base rate then amplifies that to the +0.90 DAF figure, so roughly half the DAF headline is a reporting denominator and only the precision-point interaction should be read as the effect size. The precision advantage itself does not arise because hull-margin confidence is intrinsically sharper in oxide chemistry — its within-oxide rank-AUROC is if anything the lowest of the six strata — but because oxide formation energies are where UIP predictions disagree most with the (least trustworthy) MP2020 labels, by a wide margin (RMSE

~ 0.6 versus ~ 0.09 eV/atom elsewhere), so the largest share of oxide stable-calls are false positives and confidence-based abstention has the most precision to recover. Both compounding factors — the false-positive headroom and the base-rate denominator — therefore point at the same place: the chemistry where the DFT reference is weakest. That is why we read the interaction as a diagnostic of reference-data trustworthiness rather than a chemistry-specific abstention policy. Intermetallics, by contrast, contribute few CI-firing cells for MACE and ORB, whose intermetallic gain is statistically indistinguishable from zero.

b. The oxide anchoring is not manufactured by WBM’s construction. A natural worry is that the effect is an artifact of how the WBM test set is built — elemental substitution into known prototypes² — rather than a property of UIP reliability. Three features of the analysis rule this out. First, the interaction is measured in DAF (precision normalized by each stratum’s own stable base rate) under a matched-yield control that fixes the same absolute number of surfaced candidates in every stratum, so neither the abundance of oxides in WBM (they are $\sim 10\%$ of the joined set, not a dominant class) nor their base rate can mechanically inflate the oxide gain — both quantities are divided or held out. Second, the driving quantity is the UIPs’ formation-energy error, which is $\sim 7\times$ larger in oxides (Table II) and appears at near-identical magnitude across four independently trained architectures; a sampling artifact of WBM would not reproduce the same oxide-error signature across unrelated models, whereas a genuine transferability gap on oxide chemistry would. Third, the effect survives controls that a construction artifact should not: dropping oxygen entirely still fires the interaction (37/40 cells), the metal-fraction and electronegativity stratifications — which never reference oxide membership — reproduce it, and the time-like WBM-round splits preserve its direction. The oxide anchoring therefore reflects where UIPs are least accurate, not how WBM was assembled.

c. Why UIP-label disagreement is largest in oxides — and why the oxide labels are least trustworthy there. The diagnostics establish that the disagreement between a UIP’s predicted formation energy and its MP2020-corrected training label is largest in oxides (Table II). We are careful about the level of this statement. The UIPs are trained on the already-MP2020-corrected labels^{3,9}, so their residual is a model fitting error on corrected targets, not a leakage of the semilocal-DFT (GGA) error into the prediction — that GGA error lives in the labels and is corrected before the model ever sees it. Two documented properties of universal potentials make oxides the hardest chemistry to fit: a systematic potential-energy-surface softening / energy underprediction shared

across uMLIP architectures¹⁰, and a per-chemistry error concentration in oxide and transition-metal chemistry reported in independent assessments¹¹. The MP2020 corrections are themselves composition- and oxidation-state-dependent step functions⁹ that a smooth interatomic potential is ill-suited to reproduce, which plausibly amplifies the oxide fitting gap (the anomalously large oxide formation-energy RMSE in Table II, several times the models’ aggregate error, is consistent with such a hard-to-fit seam and is not separately isolated here). Compounding this, the oxide labels are also the least reliable slice of the ground truth: correlated-*d* transition-metal oxides require a Hubbard-*U* treatment that the PBE-based Materials Project reference does not consistently carry^{12,13}, and the very existence of the MP2020 anion/oxidation-state corrections is a symptom that oxide DFT energetics carry uncorrected seams. Two things therefore co-locate in oxides — the chemistry a UIP fits worst, and the chemistry whose reference energies are least trustworthy — and our disagreement metric cannot separate them. The abstention benefit is present in main-group oxides too (Table V), so it is not carried by the correlated-*d* sub-family alone; but we do not claim to have isolated a single physical cause. The demonstrated result remains the two-factor account of the abstention benefit above (error headroom $\times$ base-rate normalization), now read as headroom in UIP-label disagreement rather than in UIP error against a trusted truth.

d. The oxide benefit is not a transition-metal-oxide artifact. Splitting the oxide stratum into transition-metal oxides (formula containing a *d*-block metal; 17,369 structures) and main-group oxides (7,651) shows the matched-yield abstention benefit is present in both sub-families for all four UIPs (Table V): keeping the more-confident half of each sub-family’s called-stable structures gives a positive DAF gain in every (model, sub-family) cell. Main-group oxides, which lack the transition-metal *d*-electron localization seam, show a comparable benefit, so the effect is not carried by transition-metal oxides alone — consistent with the anion-level (oxygen) DFT correction being the broad driver of the oxide error (post-hoc descriptive; full values in the reproduction archive).

e. What DFT verification would and would not add. A natural question for a screening paper is whether any prediction was verified against density-functional theory. For this audit the question is answered by construction: the estimand throughout is agreement with WBM’s MP2020-compatible DFT ground truth^{1,9}. Every structure a UIP calls stable, every abstention, and every stable/unstable flip is scored against an already-computed DFT hull distance, so the precision, DAF, and coverage numbers are DFT-referenced by definition. Running new DFT on the flipped or abstained candidates would not test any claim made here; it would test the fidelity of the WBM ground truth itself, which

TABLE V: Within-oxide split (post-hoc descriptive): matched-yield abstention detail for transition-metal oxides ($n = 17,369$) versus main-group oxides ($n = 7,651$), per UIP. n called-stable is the sub-family’s called-stable count at the loose budget; precision is the true-stable fraction among called-stable structures, full budget $\rightarrow$ confident half. DAF gain is positive in all eight cells.

UIP	Sub-family	n called-stable	Precision (full $\rightarrow$ half)	DAF gain
CHGNet	TM-oxide	1778	0.771 $\rightarrow$ 0.917	+1.24
	Main-group	918	0.733 $\rightarrow$ 0.867	+0.97
M3GNet	TM-oxide	2907	0.512 $\rightarrow$ 0.579	+0.56
	Main-group	1292	0.632 $\rightarrow$ 0.785	+1.11
MACE	TM-oxide	2221	0.757 $\rightarrow$ 0.825	+0.58
	Main-group	1047	0.860 $\rightarrow$ 0.973	+0.83
ORB	TM-oxide	2003	0.878 $\rightarrow$ 0.950	+0.61
	Main-group	962	0.922 $\rightarrow$ 0.938	+0.11

is upstream of our contribution and shared by every Matbench Discovery result. We therefore run no new DFT, and none is required for the audit’s validity: unlike a discovery study that must confirm novel predictions, our claims are statements about decision behavior measured against DFT labels that already exist. One caveat follows directly, and bounds the oxide reading: because the estimand is agreement with those labels, the audit inherits their reliability, and — as established above (“Why UIP–label disagreement is largest in oxides”) — the oxide labels are the benchmark’s least reliable slice of the ground truth. The oxide-anchored abstention benefit is therefore anchored to the slice of WBM where the ground truth is weakest, so the finding is best read as “UIP–label disagreement is largest, and least trustworthy, in oxides,” not as a certified statement about UIP error against a fully trusted truth. This does not affect the non-oxide interactions or the method itself, only the interpretation of the oxide anchor. To be explicit, our claims are statements about decision behavior measured against DFT labels that already exist.

f. Why this survives where the “beats margin” wedge did not. The companion study tested whether a learned or calibrated confidence signal (isotonic p_{stable} , cross-model disagreement) beats hull margin at matched coverage and matched stable-call yield, and found that none did (0/12 cells): margin was undefeated, not shown insufficient. The present result does not contradict that; it asks

a different, orthogonal question — whether margin’s own well-established abstention benefit is uniform across chemistry, and finds it is not. Pooling oxides (large benefit) with intermetallics (no benefit) at matched yield washes the average out, so the signal lives in the interaction across anion strata, not in a main-effect claim about whether margin beats some other baseline, which is why it survives the same common-absolute-yield control that companion study used. We carried that control through genuine anion-class strata, leave-one-element-out, and WBM-round splits, and it survives all of them with a clean shuffle-null.

g. Relation to certified and ensemble-uncertainty approaches. Two adjacent literatures address confidence for UIPs, but neither reports a chemistry-conditioned value of abstention. Proof-Carrying Materials⁸ delivers a global, non-stratified risk-of-failure model (AUC-ROC 0.938) backed by formal safety certificates — a calibration/certification contribution, not a stratum $\times$ coverage DAF interaction, and it reports no chemistry-conditioned abstention benefit. Multi-head-committee ensemble uncertainty¹⁴ instead constructs a richer confidence signal (committee-of-heads prediction standard deviation) and validates it via force-error correlation and active-learning data efficiency; it neither uses the free $|\hat{e}_{\text{hull}}|$ baseline nor tests whether the value of any confidence signal for stability screening varies by chemistry. Recent calibration-focused work on UIP uncertainty — flexible post-hoc calibration¹⁵ and uncertainty quantification for *misspecified* potentials¹⁶ — sharpens the confidence estimate itself but likewise does not condition the value of abstention on chemistry. Closest in framing, PROBE¹⁷ recasts UIP reliability as *selective classification* via a post-hoc probe on frozen representations, but does so at the per-atom level during molecular dynamics rather than at the WBM stability decision, and reports no chemistry-conditioned value of abstention. The present work is orthogonal to all three: it is deliberately agnostic to the source of the confidence signal — the oxide-anchored interaction persists whether the signal is the cheapest single-model $|\hat{e}_{\text{hull}}|$ or a four-model committee variance (Figure 6, Table III) — and isolates a stratum-dependent abstention benefit that neither the certification nor the ensemble-uncertainty literature reports. In Digital Discovery, Yu¹⁸ similarly probes the reliability limits of uncertainty-aware screening — there, uncertainty-guided active learning for lead-free halide perovskites, where the confidence signal recurrently flags heavy *d*-electron halides as likely functional-driven false positives. Our matched-yield audit occupies the same reliability-of-screening lane at benchmark scale (32 UIPs, $\sim 257\text{k}$ structures): rather than choosing which candidates to acquire, it asks whether the value of abstaining is itself chemistry-dependent, and — consistent with that work’s caution about

chemistry-localised failure — finds its largest payoff precisely in the oxide stratum.

h. Statistical-rigor and reproducibility precedent in the data-centric methodology literature. The audit-protocol framing here — a frozen, provenance-gated re-analysis of already-published prediction files rather than a new trained model — has direct precedent in the data-centric materials-ML methodology literature. Persaud et al.¹⁹, in Digital Discovery, show that even a single published materials-ML result can fail to reproduce without careful provenance tracking, the failure mode our md5-gated, byte-identical reproduction is built to rule out. Ottomano et al.²⁰, likewise in Digital Discovery, show that a common materials-informatics assumption — that aggregating more training data uniformly helps — does not survive a matched, controlled comparison, the same matched-yield discipline this paper applies to abstention rather than to training-set size. Witman and Schindler’s MatFold²¹, also in Digital Discovery, standardizes cross-validation splitting for materials-discovery models precisely because naive splits bias performance estimates, echoing why this paper reports LOEO and WBM-round splits as robustness gates rather than a single held-out number. On the confidence-signal side, a perspective in Digital Discovery on uncertainty in ML atomistic modeling²² surveys Bayesian and ensembling UQ approaches broadly, and architecture-search-based ensembling for molecular-property UQ²³, also in Digital Discovery, offers a richer alternative confidence signal in the same spirit as the committee-variance comparison here (Figure 6); neither reports a chemistry-conditioned abstention value, the gap this paper fills. Within *this* journal specifically, the closest genuine anchor is Beck et al.’s¹⁴ multi-head-committee uncertainty quantification for UIPs (discussed above), which validates a richer confidence signal via force-error correlation and active-learning data efficiency but — like the Digital Discovery methodology literature above — does not test whether the value of any confidence signal for stability screening is itself chemistry-conditioned.

i. Scope and what stays dead. We claim an abstention-quality interaction at matched yield, read as a diagnostic of where the reference data is least trustworthy — and nothing beyond that diagnostic. The positive deliverable is that reading, not a forward-looking policy: stratification does not re-order the four models’ overall stability-screening performance — the rank-reordering framing tested earlier remains dead — we do not assert per-round significance for every model, and (per Threats to validity and the falsified knapsack) we do not claim a deployable stratum-targeting allocator. The two under-powered WBM rounds (4 and 5) are reported transparently as a power limitation concentrated in the weakest firer (ORB), not as evidence of temporal decay.

j. Fixed-hull recompute: validated. The predicted hull distance used throughout is the fixed-hull reconstruction $\hat{e}_{\text{hull}} = e_{\text{hull}}^{\text{true}} + (\hat{e}_{\text{form}} - e_{\text{form}}^{\text{true}})$, which holds the true convex hull fixed and moves only by the model’s formation-energy error. We validate this shortcut against a from-scratch convex-hull recompute off the Materials Project phase diagram on a stratified 5,000-structure subset (4,998 evaluated). Once the MP reference surface is placed on the same MP2020-corrected formation-energy convention as the query (the correction layer that built the ground-truth column; a first execution mistakenly used total DFT energies, and its implausible multi-eV apparent deltas were what flagged the mismatch — a useful audit tell, since the delta $\hat{e}_{\text{hull}}^{\text{fixed}} - \hat{e}_{\text{hull}}^{\text{recomp}}$ is model-independent by construction, so a large value can only be a reference bug), the recompute reproduces the published $e_{\text{above hull}}$ to $\sim 10^{-3}$ eV/atom in every stratum (per-stratum mean $|\Delta| \leq 0.0009$) and changes the stable/unstable call for only 13 of 4,998 structures (0.26 %; CHGNet 5, M3GNet 6, MACE 1, ORB 1). Those 13 flips are confined to the mixed-anion strata (halide 5, pnictide 4, chalcogenide 3, other 1), with zero in oxide and zero in intermetallic — so the two strata that anchor the headline interaction are exactly where the fixed-hull call is most stable. One scoping caveat: this recompute certifies the fixed-hull *reconstruction* — that recomputing $e_{\text{above hull}}$ from each model’s predicted formation energy against the true Materials Project reference set reproduces the shortcut — and not the deeper approximation that the competing phases defining the hull are themselves unmoved by a UIP’s errors on those phases (every reference entry here is DFT ground truth, not a UIP prediction). The stability calls the paper actually uses are the reconstructed ones, and it is those that are validated: the fixed-hull approximation is a sound basis for the stability calls throughout, not merely an acknowledged limitation.

k. Roster-wide universality of the oxide anchor. The four-UIP headline rests on cached predictions that carry the MP2020 corrections natively. To test whether the oxide anchoring is a property of those four models or of universal potentials as a class, we converted every credential-free WBM prediction file on the Matbench Discovery Figshare record (version 25): 44 files — 27 trained on Materials Project²⁴ data only (the 2022–24 “MP generation,” including seven 2020-era classical baselines) and 17 trained on the newer OMat24²⁵ / Alexandria²⁶ / OpenLAM corpora (the “OMat generation”). Each file’s MP2020-corrected formation-energy column is the authoritative published prediction — byte-identical to the frozen CHGNet anchor over all 256,963 rows — so the expansion sits on the same ground truth as the headline, not a re-derivation. All 44 raw sources were re-verified against their Figshare-published md5 checksums (an automated verification pass,

44/44 OK). The roster and its two-arm generational aggregation follow a pre-specified analysis plan written before the run; it is not git-timestamped ahead of the run, unlike the alternative-stratification plan below, so we label it a pre-specified plan rather than a preregistration.

Model families represented in the roster. Beyond the four headline architectures already introduced — CHGNet³, M3GNet⁴, MACE⁵ (here the MACE-MP-0 foundation checkpoint²⁷), and ORB⁶ — the 32-model roster (Table VI) spans several further architecture families: message-passing equivariant networks (NequIP²⁸, its strictly local variant Allegro²⁹, and the equivariant transformer EquiformerV2³⁰), the graph-generalized atomic cluster expansion family (GRACE³¹), the DPA line of large atomic models (DPA-2³² and DPA-3³³), and two recent smooth/local-frame potentials (eSEN³⁴ and AlphaNet³⁵), alongside MatterSim³⁶, Orb-v3³⁷, and SevenNet^{38,39} introduced above. The oxide anchor’s universality across this architecturally diverse set (Table VI) is part of what licenses reading it as a property of universal potentials broadly rather than of any one model family.

One published file is a non-independent duplicate. Of the 44 converted files, the OMat-generation Orb-v2 file (Figshare file 51607307, “orbff-v2-20241011.csv.gz”) is byte-identical ($\max |\Delta e_{\text{form}}| = 0$ over all 256,963 rows) to the frozen MP-generation “orb” headline anchor: both trace to the same published Orb-v2 checkpoint. The frozen anchor had been labelled “Orb-v2 MPtrj-only,” but re-fetching and cross-checking both published Orb-v2 files shows the anchor matches the general Orb-v2 checkpoint (file 51607307) exactly and the true MPtrj-only checkpoint (file 51607310, “orbff-mptrj-only-v2-20241014.csv.gz”) only approximately (mean $|\Delta| = 0.037$, max $|\Delta| = 1.99$ eV/atom). This file is therefore excluded from the generational contrast as a non-independent duplicate rather than an additional OMat-generation model, leaving 43 independent models. This is a labelling/duplication issue, not a corrupt or fabricated file: its md5 gate passes cleanly, and it is the only one of the 44 sources that is non-independent.

Correction to the earlier Orb-v3 “dissenter” claim. A previous version of this check self-inferred predictions for a four-model roster and reported Orb-v3 as a sharp oxide-anchoring reversal (-1.83). That figure was an artifact of the self-inference, which omitted the MP2020 anion/oxide corrections and thereby imprinted a systematic anion-count-proportional offset ($\approx +0.11$ – 0.59 eV/atom, largest on oxides) that the ground truth and the cached CSVs both carry, leaving the reconstructed-hull oxide calls degenerate. On the *authoritative published* Figshare v25 predictions, Orb-v3’s oxide-vs-halide abstention interaction is $+0.64$ (95 % CI $[0.52, 0.76]$) — positive and oxide-anchored like every other modern potential. We correct the earlier reversal explicitly: it was a harness artifact, not a

model property.

The oxide anchor is universal across the modern-UIP roster. Under the pre-specified inclusion rule (≥ 50 called-stable structures per stratum and $\geq 99\%$ WBM coverage, which keeps the common-yield budget non-degenerate), 32 modern UIPs qualify (16 MP-generation, 16 OMat-generation, after dropping the duplicate). The oxide-vs-halide matched-yield interaction is positive with a prototype-blocked 95 % CI excluding zero for all 32 (16/16 MP-generation, 16/16 OMat-generation), and oxide is the single highest-benefit stratum in 29/32 (the three exceptions are MP-generation models whose arg-max is the residual “other” stratum, not a reversal); per-model interactions range from +0.31 to +1.31 (Table VI). The pre-specified hypothesis that the newer OMat generation would dissolve or reverse the anchoring is not supported: a model-level permutation test on the generation label gives $p = 0.68$ (observed MP–OMat gap in mean interaction -0.037 , 95 % CI $[-0.21, 0.13]$; $n_{\text{MP}} = 16$, $n_{\text{OMat}} = 16$). We read this as no detectable generational difference in the oxide anchor: the CI is wide relative to the observed gap and the analysis is underpowered (at $n = 16$ per arm) to exclude a moderate-magnitude generational shift, so “generation-independent” should be read as an honest null within this roster’s power, not as a demonstrated invariance.

Classical baselines and multiplicity. The seven 2020-era classical baselines (CGCNN, CGCNN+P, MEGNet, Wrenformer, Voronoi-RF, ALIGNN, BOWSR-MEGNet; all MP-trained) are by contrast weak and mixed — five carry a positive CI-excluding oxide-vs-halide interaction but oxide is the arg-max stratum for only two, and Voronoi-RF reverses it (-0.12 , CI $[-0.22, -0.02]$) — so the effect is specific to modern universal potentials, not shared by every ML formation-energy model. Over the full included-UIP interaction family ($32 \times 15 = 480$ cells) 378 CI-exclude zero and 369 survive Benjamini–Hochberg FDR⁴⁰ at $q = 0.05$; the oxide-vs-halide subfamily is 32/32 under both, with a clean within-stratum confidence-shuffle null (0/480). The Holm–Bonferroni family-wise procedure⁴¹ survives 0/480 (and 0/32 on the oxide-halide subfamily) at $\alpha = 0.05$: this is a bootstrap-resolution limit, not evidence against the effect — the minimum achievable two-sided bootstrap p at $n_{\text{boot}} = 1000$ is ≈ 0.002 , and Holm’s family-wise correction over 480 (or 32) cells demands $p < 0.05/480 \approx 1.0 \times 10^{-4}$ (or $0.05/32 \approx 1.6 \times 10^{-3}$), both below the achievable floor; the BH-FDR and raw-CI readings above are the informative ones here, mirroring the treatment of the equivalent Stage-1 Holm result. Numbers trace to the roster-expansion audit in the reproduction archive (the corrected re-run; supersedes the contaminated earlier run, retained with a provenance note).

Two published files with non-physical MAE, retained with disclosure. Two MP-generation files as published on Figshare v25 — GRACE-2L-MP|\hat{e}_{\text{form}}| up to 10^{22} and 10^{33} eV/atom) that make their naive mean absolute error non-physical (5×10^{16} and 10^{28} eV/atom); trimming those rows restores physical MAEs of 0.057 and 0.046 eV/atom. Because the analysis ranks and thresholds predictions rather than averaging them, these rows do not affect the result: removing them shifts each oxide-vs-halide interaction by ≤ 0.013 (well inside its ~ 0.3 -wide CI) and leaves the arg-max stratum and CI-exclusion unchanged, so both models are retained among the 32 with this disclosure rather than silently trimmed.

Frozen-anchor provenance disclosure. Three of the four headline anchors (M3GNet, MACE-MP-0, and the “orb” anchor discussed above) are frozen on-disk snapshots that are not the current Figshare v25 publication: cross-checking each against its re-downloaded v25 counterpart shows measured drift (mean / max absolute difference in eV/atom: M3GNet 0.036/3.65; MACE-MP-0 0.050/349.59; orb 0.037/1.99; CHGNet is byte-identical, 0/0). The drift does not change any reported number — the four-UIP headline and the roster expansion both use the frozen snapshots (or, for orb, the published file the snapshot equals) consistently, and the drift magnitudes are disclosed here for provenance rather than folded into any statistic.

l. Threats to validity. The primary confound is surfaced-candidate yield, addressed by the common-absolute-yield budget. Stratum-definition sensitivity is probed two ways, both of which survive: by the LOEO of the anion-formers (F, S, N, P) and the drop-oxygen run, and — directly addressing the “cherry-picked stratifier” objection — by re-running the entire matched-yield interaction under two alternative stratum definitions that do not use the anion-priority ordering, preregistered ex ante and git-timestamped (2026-07-02 10:52 UTC, before the analysis runs at 11:01 UTC): amount-weighted mean-Pauling- electronegativity quintiles and amount-weighted metal-fraction bins. Both reproduce the chemistry-dependence (overall verdict BOTH_ROBUST): the interaction CI-excludes zero in 31/40 (electronegativity) and 24/40 (metal-fraction) model $\times$ stratum-pair cells, with the more anion-/oxide-rich, less-metallic strata carrying the higher abstention benefit. We are careful not to overstate the electronegativity stratifier as a clean anti-cherry-pick rebuttal: of its 31 CI-excluding cells only 6/40 fire *in the hypothesized anion-rich>metal-rich direction* (the rest exclude zero in mixed or opposite directions, because its top quintile pools oxides and halides — opposite-extreme anion classes), whereas metal-fraction reproduces the direction

TABLE VI: Roster-wide oxide-vs-halide matched-yield abstention interaction I (prototype-blocked 95 % CI) for all 32 included modern UIPs, by training generation (models are Matbench Discovery leaderboard identifiers; the published Orb-v2 OMat file is excluded here as a non-independent duplicate of the MP-generation “orb” anchor, text). $I > 0$ with CI excluding zero for all 32; oxide is the highest-benefit stratum in 29. GRACE-2L-MPtrj and SevenNet-0 carry non-physical published MAE from a few outlier rows that do not affect this rank-based interaction (text).

MP generation ($n = 16$)			OMat generation ($n = 16$)		
model	I	95 % CI	model	I	95 % CI
eqv2_s_mp	+1.16	[1.02, 1.29]	allegro_oam	+1.31	[1.19, 1.44]
matris_mptrj	+1.01	[0.88, 1.13]	dpa31_ft	+1.13	[1.00, 1.25]
dpa3_v2_mptrj	+0.99	[0.86, 1.13]	dpa3_v2_openlam	+0.98	[0.86, 1.11]
allegro_mp	+0.95	[0.80, 1.11]	grace_1l_oam	+0.91	[0.77, 1.04]
grace_2l_mptrj	+0.91	[0.76, 1.06]	eqv2_m_omat	+0.90	[0.77, 1.03]
mattersim	+0.83	[0.71, 0.94]	dpa3_v1_openlam	+0.83	[0.71, 0.96]
dpa3_v1_mptrj	+0.83	[0.70, 0.95]	alphanet_oma	+0.70	[0.58, 0.82]
hienet	+0.74	[0.60, 0.87]	esen_oam	+0.70	[0.58, 0.81]
nequip_mp	+0.59	[0.45, 0.73]	nequip_oam_xl	+0.67	[0.55, 0.77]
matris_10m_mp	+0.58	[0.47, 0.69]	orb_v3	+0.64	[0.52, 0.76]
eqnorm_mptrj	+0.50	[0.37, 0.63]	mace_mpa	+0.62	[0.48, 0.75]
alphanet_mp	+0.49	[0.37, 0.61]	grace_3l_oam	+0.57	[0.45, 0.67]
sevennet_0	+0.48	[0.34, 0.61]	grace_2l_oam	+0.55	[0.44, 0.68]
nequix_mp	+0.42	[0.29, 0.57]	nequip_oam	+0.54	[0.42, 0.64]
sevennet_13i5	+0.41	[0.27, 0.55]	matris_oam	+0.50	[0.40, 0.61]
esen_mp	+0.38	[0.25, 0.52]	grace_2l_oam_l	+0.31	[0.19, 0.42]

cleanly (14/40 in-direction). It is therefore metal-fraction, not electronegativity, that provides the clean directional reproduction. Both nonetheless survive within-family Holm correction (5 and 13 anion-consistent survivors, Holm-adjusted p down to 8.0×10^{-3}) with a clean shuffle-null (0/40 each). Metal-fraction is cleanly monotonic (mf1, nonmetal-rich, highest benefit; mf5,

96 % intermetallic, lowest); the electronegativity quintiles confirm the heterogeneity but are non-monotonic at the top quintile for a chemically sensible reason — it pools oxides and halides, the two opposite-extreme anion classes — reported transparently in a dedicated alternative-stratification findings note. The chemistry-dependence is therefore not an artifact of the single anion-priority classification rule. Thin-stratum DAF instability is guarded by minimum-row and minimum-called-stable thresholds. Multiple comparisons are handled three ways: by requiring a majority of cells rather than any single cell, by pairing every real test with a matched shuffle-null (zero false positives observed), and by a formal correction: per-cell two-sided bootstrap sign p -values with Benjamini–Hochberg at $q = 0.05$ applied within each family retain 46/60 Stage-1, 539/700 LOEO, and 135/300 WBM-round cells (shuffle families: 0/60, 0/700, 0/300), and the family-wise Holm–Bonferroni procedure at $\alpha = 0.05$ retains 40/60 Stage-1 cells in a higher-resolution 10^4 -bootstrap pass (Holm is bootstrap-resolution-limited at $m = 700/300$ with 10^3 draws, so BH-FDR is the formal correction reported for the robustness families). One structural caveat applies to all of these corrections: within a model the 15 stratum-pair contrasts are not independent — with six strata they span only five independent dimensions, since any pair difference is fixed by the others (e.g. (oxide – halide) = (oxide – intermetallic) – (halide – intermetallic)). BH and Holm treat the linked contrasts as separate tests, so the effective number of independent stratum comparisons is five per model, not fifteen; the corrected counts should be read with that redundancy in mind. Every reproduced cell in these correction runs was asserted identical (median, CI, exclusion status) to the frozen Stage-1/Stage-2 result JSONs before p -values were attached. The single confidence signal $|\hat{e}_{\text{hull}}|$ is the free hull-margin baseline; richer signals such as cross-model ensemble disagreement are future work. The companion study’s audit found disagreement does not survive the joint matched-coverage-and-yield control and can underperform margin there (e.g. CHGNet cov_50 matched-yield gain -0.0724 , 95 % CI $[-0.0845, -0.0610]$, excluding zero in the negative direction), so whether a richer signal would strengthen or dilute the stratified interaction is an open question, not an assumed one. The tight/loose yield split itself ($Y_{\text{tight}} = Y_{\text{loose}}/2$) was fixed once, post hoc; a subsequent sensitivity sweep re-ran the identical matched-yield machinery at $Y_{\text{tight}} = \text{round}(Y_{\text{loose}} \cdot r)$ for $r \in \{1/3, 0.4, 0.5, 0.6\}$ (same seed, same 1000-bootstrap CIs). The interaction count is stable across the grid — 45/60, 47/60, 48/60, and 47/60 cells CI-excluding zero at $r = 1/3, 0.4, 0.5, 0.6$ respectively, with the shuffle-null clean (0/60) at every ratio — and the headline CHGNet oxide-vs-halide cell fires at all four ratios (median interaction $+1.14, +0.99, +0.90, +0.75$; monotone in r as expected, since a looser tight budget dilutes the abstention contrast).

The ratio choice is therefore a confirmed non-issue for the qualitative claim, though the interaction magnitude scales with the budget ratio. A prospective, temporally held-out check calibrates the stratified-abstention policy on early WBM acquisition rounds and applies it to later rounds; it shows the oxide-highest ordering does not transfer out-of-time, and we report both specifications per the forking-paths rule under which the arm was run. In the primary pooled specification (calibrate rounds 1–3, apply to the pooled later rounds 4–5), the oxide abstention benefit stays positive for all four UIPs but is the top-benefit stratum for none (0/4) and falls below the mean of the other strata for all four (0/4 beat the other-strata mean; overall verdict MIXED); the high-benefit strata out-of-time are intermetallics and pnictides. In the round-5-only sensitivity specification (calibrate rounds 1–4, apply to round 5) the picture is weaker still: ORB’s future-round oxide gain is marginally negative (−0.001) and oxide beats the other-strata mean for only 2 of 4 models. The oxide-anchored ordering is therefore a property of the pooled evaluation and does not survive a prospective, temporally held-out test; whether a stratified policy should target oxides in deployment is consequently not established here.

m. Next step — and a caution on turning the interaction into an allocator. The tempting policy reading — “spend the abstention budget preferentially on oxide candidates rather than applying a uniform global confidence cutoff” — does not follow automatically, and our own follow-up falsified its strongest form. A constrained greedy knapsack allocator that reallocates a fixed total-coverage budget across strata by marginal DAF (coverage 0.7, seed 20260629, 1000-bootstrap CIs) failed to CI-dominate the global threshold on per-stratum DAF: of 24 (model × stratum) cells, 5 improved with CI excluding zero but 14 worsened with CI excluding zero, and the equal-weight aggregate mean DAF improved with CI excluding zero in 0 of 4 models. The mechanism is a conservation constraint: at matched total coverage, per-stratum DAF depends only on that stratum’s kept count, so any reallocation is zero-sum across strata — what oxides gain, other strata give back. The interaction documented here therefore licenses stratified reporting and stratified expectations (which chemistries a confidence cutoff will actually help, and by how much), and it licenses spending additional abstention capacity where it pays, but it does not license reshuffling a fixed budget away from a global threshold. The alternative-stratification robustness reported above (Threats to validity) already addresses the stratifier-choice objection directly; a richer-confidence comparison remains for a benchmarks-track submission.

AUTHOR DECLARATIONS

Conflict of interest

The author has no conflicts to disclose.

Use of AI tools

AI-based coding and writing assistant tools (large language models) were used to assist with drafting and editing of the manuscript text and analysis code; the author reviewed and verified all content and takes full responsibility for it.

Author contributions

Zeyu Fu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing.

DATA AVAILABILITY

All inputs are the public Matbench Discovery¹ on-disk cached artifacts (WBM summary ground truth from² and the four cached UIP prediction CSVs); no authentication or run-time download is required. The genuine ex-ante commitment — the general strata direction and kill criteria — is a dated pre-analysis design memo (written before Stage-1; committed to the version-controlled repository in the pre-push baseline, with the other previously-untracked frozen artifacts). The Stage-2 analysis-protocol log is, by its own disclosure, a same-session analysis protocol and results log rather than an ex-ante preregistration: it records that the exact success thresholds and budget parameters were fixed during the Stage-1/Stage-2 runs themselves. The one preregistration with a git-verifiable ex-ante ordering is the alternative-stratification prereg, whose commit precedes its result commit (see Threats to validity). The frozen results are the Stage-1 matched-yield interaction/shuffle-null record and the Stage-2 LOEO/WBM-round robustness record, each with a SHA-256 manifest over inputs, script, and output. Supplementary frozen records cover the budget-

ratio sweep, the formal multiple-comparison correction, the falsified stratum-aware allocator, the two alternative-stratification robustness runs with their cross-stratifier multiplicity summary, and the corrected insurance arms (the fixed-hull validation and the modern-roster reproduction), each with its own manifest. The prototype-blocked Stage-1 CIs resample WBM AFLOW prototype clusters, and the mechanism-extension post-hoc diagnostics — the within-oxide transition-metal/main-group split (Table V) and the prospective temporal-transfer specifications (pooled and round-5-only; Table II and the practitioner payoff) — are each labelled post-hoc descriptive. The code is developed at GitHub (<https://github.com/PeterPonyu/materials-mlip-research>) and archived at Zenodo under DOI 10.5281/zenodo.21130295; this DOI is currently reserved on a draft deposition and becomes active when the record is published. A single self-contained figure pipeline — one R/ggplot2 script per figure, following the shared portfolio paper-template pattern — reads those frozen result records directly and regenerates every figure; no value in this paper is hand-entered. (Two earlier figure pipelines were quarantined after an audit found a duplicated R and matplotlib pipeline silently overwriting each other’s outputs; the present single R pipeline supersedes both.) The analysis scripts, frozen result records, manifests, figure scripts, and this manuscript are committed to a version-controlled GitHub repository (<https://github.com/PeterPonyu/materials-mlip-research>) and archived at Zenodo under DOI 10.5281/zenodo.21130295, consistent with AIP Publishing’s Research Data Policy template “data openly available in a public repository that issues datasets with DOIs,” alongside an actively maintained development mirror. The GitHub repository is private during peer review (a portfolio-wide precaution against scooping) and will be made public on acceptance; the Zenodo DOI is currently reserved on a draft deposition and becomes active on publication.

n. Licensing, compute, and packaging. The entire data-and-model chain is license-clean with no non-commercial clause (Table VII). All analysis — including the roster expansion (44 published-potential files, 43 independent models) — is CPU/pandas only and reads published, credential-free Matbench Discovery predictions (no network access, tokens, or model weights at run time). A superseded earlier arm self-inferred four models on a single RTX 4090D (24 GB; measured peak GPU memory 6.3 GiB, peak utilisation 84 % over the 256,963-structure pass, per-minute `nvidia-smi` log); it is retained only as the diagnostic that surfaced the MP2020-omission artifact corrected in the roster analysis above, and contributes no reported number.

TABLE VII: License provenance for the data and models. No component carries a non-commercial (CC-BY-NC) restriction.

Component	Role	License
WBM test set ²	ground-truth stability	CC-BY-4.0
Matbench Discovery code ¹	benchmark/eval harness	MIT
MACE / MACE-MP-0 ⁵	UIP (code + checkpoint)	MIT
MatterSim ³⁶	UIP (code + checkpoint)	MIT
Orb-v3 ³⁷	UIP (code + weights)	Apache-2.0
SevenNet-MF-ompa ^{38,39}	UIP (code + checkpoint, $\geq 0.12.0$)	MIT

o. Reproducibility-chain status (self-reported). Both Stage-1 and Stage-2 have now been independently re-executed on CPU from the same on-disk cached inputs and their manifests regenerated against the current scripts. The Stage-1 manifest reproduced its frozen output byte-for-byte identical across two independent re-runs in different Python/numpy/pandas environments. The Stage-2 manifest, whose recorded script hash had previously gone stale relative to the on-disk Stage-2 analysis script (the script file was touched after the manifest was first generated), was regenerated by this re-run: the script hash now matches the on-disk script, and the frozen output again reproduced byte-for-byte identical, confirming the Stage-2 result is genuinely deterministic and unaffected by the earlier hash staleness.

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